

Hierarchical Multi-Agent Swarm Algorithm for Optimized Hospital Resource Allocation over Wireless Networks

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Abstract

In a smart hospital environment, an effective allocation of diverse hospital resources (devices, nurses, beds, etc.) is a key issue, and it is becoming much more complex due to the rise of service demand from patients, the heterogeneity of hospital medical resources, and the extensive application of wireless healthcare technology. Moreover, the traditional methods have limitations in scalability, late decision-making, and a lack of consideration for wireless communication environments. In order to overcome these drawbacks, this paper proposes a Hierarchical Multi-Agent Swarm Algorithm (HMSA) for optimized hospital resource allocation over a wireless network. In the proposed scheme, a hierarchical multi-agent mechanism combined with a swarm intelligence algorithm is applied to distributed, adaptive, communication-aware hospital resource allocation. The multi-agent architecture comprises 4 layers, which are the healthcare device, local agent, regional coordination, and global optimization layers. These agents cooperatively observe the status of available hospital resources and healthcare services requirements. Also, a wireless-aware fitness function is designed to jointly optimize the system utilization, allocation delay, throughput, PDR, and QoS of the services. A simulation has been carried out using a large-scale smart hospital environment. Comparison results demonstrate that HMSA outperforms FCFS, Round Robin, Ant

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Colony Optimization, and Particle Swarm Optimization algorithms. The HMSA achieved utilization of 94.8%, throughput of 95.4 Mbps, PDR of 98.7%, QoS of 97.1%, and allocation delay of 96 ms. With one-way ANOVA analysis, an F-value of 32.84 was produced, and $p < 0.001$ showed that significant improvements were achieved.

Keywords: Hierarchical Multi-Agent Systems, Swarm Intelligence, Wireless Healthcare Networks, Internet of Medical Things, Resource Allocation Optimization, Smart Hospitals.

1 Introduction

1.1 Background

Recent advancements in smart healthcare environments have led hospitals to evolve into extremely interconnected cyber-physical systems. In such systems, networked devices such as monitoring systems and wearable sensors are interconnected with clinical information platforms through wireless infrastructures, enabling constant exchange of medical devices and other smart systems (Dong et al., 2023). With the development of IoMT, edge computing and intelligent networking technologies, a range of intelligent control mechanisms enabling medical professionals to conduct real-time patient monitoring and to take adaptive decisions in the hospital are now available (Veerappan & Arvinth, 2025; Dong et al., 2023). Moreover, contemporary healthcare infrastructures rely heavily on distributed intelligence and automatic communication services to support essential healthcare applications in a high-reliability, low-latency and scalable way (Ismail & Kadhim, 2024).

The problem of resource allocation becomes increasingly important as, with constant admissions of new patients, emergency events, and dynamic medical needs in a multi-heterogeneous smart hospital environment, effective management of all medical facilities is not a trivial task. Intelligent resource allocation schemes have been recently considered key components of future wireless networks in terms of efficiently utilizing network resources in various large-scale mobile networks where the heterogeneous QoS requirements are considered (Zhu et al., 2025). And the future trend of integration of distributed intelligence within the healthcare resource coordination can also be realized with cloud-radio access networks or enhanced wireless networks (Alhussien & Gulliver, 2024).

Distributed machine learning and distributed intelligence have made adaptive smart healthcare infrastructures possible and allowed for more flexible responses to real-time changes in communication environments and network conditions (Ebrahimi et al., 2024). The large-scale expansion of the health care infrastructure will require intelligent resource allocation schemes to address both the networking and the operational aspects of the system.

1.2 Research Problem

Resource Allocation: Despite advancements in wireless healthcare technologies, efficient hospital resource allocation continues to be an unresolved issue. Older scheduling systems are centralized, which has problems in addressing scalability issues where the number of patients, medical equipment, healthcare personnel, and even departments can grow to high numbers. However, due to sudden changes in health conditions, there can be resources lying idle in some sections, whereas there can be bottlenecks and delays in some other sectors.

When the modern hospital environments require optimal resource allocation, the decisions made can no longer consider networking constraints. The decisions have to take into account various factors such as network latency, packet loss, transmission overhead, quality of wireless link, and others

(Liu et al., 2023). The current approaches do not consider these communication aspects and are far from efficient resource allocation with regard to both efficiency and responsiveness. In a recent survey of 5G and 6G resource management, intelligent and responsive allocation frameworks that can support communication awareness strategies have been highlighted as one future direction (Liu et al., 2023). Similarly, dependable wireless healthcare infrastructure should support seamless operations and guaranteed service quality even in rapidly changing network conditions (Dhasarathan et al., 2024).

1.3 Motivation

The ever-increasing deployment of wireless medical devices, smart sensors, and edge computing systems leads to demand for the designing of decentralized resource management solutions that are effective in a challenging and demanding health care setting. Distributed intelligence is identified as a possible framework for the designing of systems that make autonomous decisions and perform adaptive optimization on a network of systems (Ismail & Kadhim, 2024). The scalability, robustness, and flexibility are ensured by the use of multi-agent architectures, wherein the resource management functions are partitioned among a number of autonomous agents.

However, swarm intelligence has already shown powerful performance for various optimization problems where coordination and scheduling are required for various issues related to the resource allocation in changing environments (Shaik et al., 2025). A multi-agent-based coordination mechanism along with swarm-based optimization techniques would significantly help in building smart healthcare systems that can adapt to the network constraints and demands of the network and are also related to recently proposed works on AI-enabled wireless networks and future healthcare communication frameworks (Meknassi Salime et al., 2025).

1.4 Contributions

The primary contributions of this study are summarized as follows:

1. A novel Hierarchical Multi-Agent Swarm Algorithm (HMSA) for dynamic hospital resource allocation over wireless health networks is introduced.
2. A hierarchical framework that is composed of local agents, coordination agents at the regional level, and optimization agents at the global level is designed to promote distributed decision-making and efficient scaling resource management.
3. A wireless aware swarm optimization framework that targets joint resource utilization and allocation delay minimization while optimizing communication overhead and network reliability is established.
4. Simulation is performed and compared to conventional allocation algorithms over a variety of hospital situations for validating the feasibility of the proposed framework.
5. The proposed HMSA approach is compared with conventional allocation methods to demonstrate improvements in performance and adaptability.

1.5 Research Objectives

1. To develop a hierarchical multi-agent architecture capable of autonomously coordinating hospital resources across wireless healthcare networks.
2. To design a swarm-based optimization strategy that incorporates wireless communication parameters into resource allocation decisions.

3. To evaluate the effectiveness of the proposed HMSA framework in terms of utilization, latency, throughput, reliability, and scalability.

The rest of the paper is structured as follows: Section 2 reviews the related work on hospital resources management, wireless healthcare networks, multi-agent system resources management, and swarm intelligence systems. Section 3 describes the system architecture being used. Section 4 details the proposed Hierarchical Multi-Agent Swarm Algorithm (HMSA). Section 5 shows the simulation setup and the experiment design. Section 6 contains the results and discussions, and finally, Section 7 provides the conclusion and future work.

2 Related Work

2.1 Hospital Resource Allocation Models

Resource management for the efficient provision of healthcare services has been a subject of significant study due to the growing needs for up-to-date healthcare facilities. Service responsiveness and resource utility have been significantly improved using mobility-aware task offloading schemes for the IoMT settings (Reffad & Alti, 2023). Autonomous-agent-based dynamic adaptive deployment systems have exhibited promising outcomes regarding workload balancing and healthcare task scheduling. Smart resource allocation techniques in fog-assisted IoT settings were later proposed, which yielded enhanced resource management while optimizing computational load and decreasing service latency (Shaik et al., 2025). Trustworthy and manageable resource provisioning techniques assisted by blockchain and intelligent optimization were introduced later for healthcare scenarios (Ali et al., 2024).

2.2 Wireless Healthcare Networks

Wireless healthcare networks are integral to facilitating real-time communication among medical devices, sensors, and healthcare information systems. The resource allocation schemes designed for UAV-supported medical networks have shown the relevance of intelligent communication management and adaptive network control (Seid et al., 2023; Zhu et al., 2025). AI-based wireless communication architectures have become indispensable in intelligent resource management of future healthcare networks (Kalaivnai, 2025; Sefati et al., 2024). Cloud-RAN-based networks are enabling improved flexibility and scalability of healthcare service provision (Alhussien & Gulliver, 2024). Secure networking-based energy-aware IoMT communication models have enhanced further the performance of wireless healthcare systems (Khedr et al., 2025).

2.3 Multi-Agent Resource Management

Much of the research in multi-agent system (MAS) technologies has focused on the decentralized decision-making and self-management of resources. Agent-based power management system design has effectively achieved distributed coordination between various health monitoring subsystems (Goswami et al., 2023). Secure information sharing and smart decision-making using agent-based multi-agent frameworks for healthcare applications have improved privacy. Recent advances in AI-based MAS reinforced learning systems Soy & Salwadkar, (2024) and Le et al., (2024) allow online adaptivity and group-wise optimization in complex real-time monitoring systems. Decentralized intelligence and collaborative learning in multi-agent systems can also be supported through distributed machine learning techniques (Ebrahimi et al., 2024). Federated collaborative intelligence frameworks achieve efficient resource management for massive distributed networks (Fateminasab et al., 2025).

2.4 Swarm Intelligence Approaches

It has been shown that swarm intelligence algorithms are well-suited for coordination and resource allocation-related combinatorial optimization problems. Optimization frameworks that combine genetic algorithms with reinforcement learning show the best performance in multi-agent task allocation problems (Alaya et al., 2025). Swarm robotics techniques are widely used in the automation of warehousing and logistics control of health care facilities. Coordination strategies based on collective motion and flocking mechanisms offer robust techniques for decentralized coordination of autonomous agents (Salim et al., 2025). Furthermore, agent-based simulation frameworks explain the efficiency of swarm-inspired cooperation and adaptive resource rivalry models (Deepika & Geetha, 2025; Yuvarani et al., 2025). Similar optimization mechanisms have also been employed in intelligent fishery management, environmental monitoring in urban areas, intelligent irrigation control, and adaptive communication networks (Egash & Teyene, 2023; Dong et al., 2023; Soy & Salwadkar, 2024).

2.5 Research Gap

Research has been carried out in various areas such as healthcare task scheduling, wireless resource management, multi-agent coordination, and swarm optimization with various views. However, most of them focus on specific aspects of the problem and fail to consider the hospital resource constraints and the wireless communication constraints simultaneously. Current research makes limited use of hierarchical coordinated structures that can achieve joint local, regional, and global decisions. Failed to provide a combined model of efficient resource utility, low communication latency, high network dependability, and wide scalability, which is important in massive smart hospitals. Thus, there is a need for a hierarchical multi-agent swarm-based framework to achieve adaptable, distributed, and wireless-awareness resource allocation for next-generation healthcare applications.

3 System Architecture

This Hierarchical Multi-Agent Swarm Algorithm (HMSA) architecture supports the concept of intelligent hospital resource allocation over wireless health networks. The architecture combines the use of IoMT-enabled medical devices, the coordination of multiple distributed agents, and swarm intelligence optimization to realize dynamic decision-making in healthcare. Instead of utilizing a centralized resource allocation system, HMSA is hierarchically designed to achieve distributed decision-making across multiple levels of coordination, which enhances scalability, reduces the communication overhead, and efficiently performs allocation. It has four layers, including a healthcare device layer, a local agent layer, a regional coordination layer, and a global optimization layer shown in figure 1.

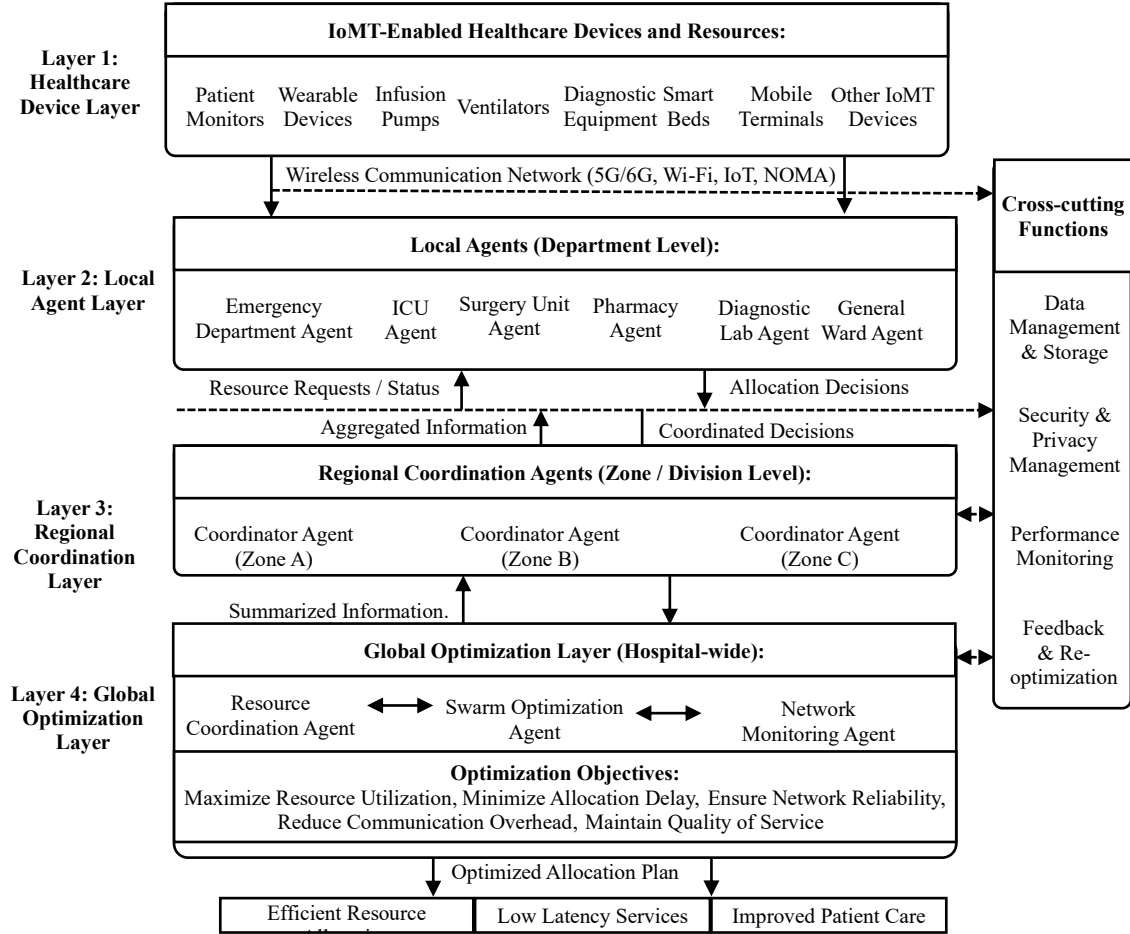


Figure 1: Hierarchical multi-agent swarm architecture for hospital resource allocation over wireless networks

3.1 Healthcare Device Layer

The Healthcare Device Layer contains a cluster of medical devices, wearable sensors, patient monitoring devices, smart hospital beds, and other IoMT devices within the hospital. These devices continuously emit operational and patient information, which is forwarded to be transmitted via the wireless network in order to aid resource monitoring and allocation. It is the main component where data is collected.

3.2 Local Agent Layer

The local agent layer includes independent agents associated with each hospital department (emergency, ICU, lab, wards, etc.). This layer controls local resource availability, current workload, and requests, performing the first level of scheduling and allocation. The local agents' decision whenever possible is made with a minimal level of communication.

3.3 Regional Coordination Layer

This is a mediating level of control coordinating a number of local agents. This layer is tasked with summarizing department information, conflict resolution, and interdepartmental resource sharing when

local resources fail. This level improves cooperation between different units of a hospital while keeping the benefits of distributed control over the healthcare environment.

3.4 Global Optimization Layer

It also contains a Resource Coordination Agent (RCA), a Network Monitoring Agent (NMA), and a Swarm Optimization Agent (SOA), which together will analyze the different assignment solutions and select the most efficient one. Resource utilization, latency of communications, reliability of network, and quality of service are analyzed at the same time in order to reach efficient hospital-wide assignment.

3.5 Wireless Communication Framework

The wireless communication structure interconnects healthcare devices, local agents, regional coordinators, and the global optimization part. It supports the continuous exchange of information, real-time observation, distributing the allocation decisions, and monitoring the communication performance, like latency, throughput, and the delivery rate of packets. Integrating communication-aware optimization, the proposed system can adjust the allocation schemes depending on the network state.

3.6 Operational Workflow of HMSA

The HMSA workflow starts with resource and patient data being obtained from medical devices. Departmental requests are analyzed and tentative allocations made by the local agent, and unmatched requests are passed on to the regional coordinator to rebalance resources. The remaining complex allocation problems are passed to the Global Optimization Layer, which utilizes the swarm-based optimization mechanism to arrive at the best allocation. The results are broadcast throughout the entire hierarchy and continuously updated.

4 Proposed Hierarchical Multi-Agent Swarm Algorithm (HMSA)

This section proposes the Hierarchical Multi-Agent Swarm Algorithm (HMSA) that integrates Hierarchical Multi-Agent coordination with Swarm intelligence optimization for efficient hospital resources allocation over wireless healthcare networks. HMSA utilizes an automatic decision-making ability at several hierarchy levels with the consideration of communication-aware optimization criteria. The algorithm perpetually monitors the availability of resources and requests of the patient as well as the network situation and allocates healthcare resources with minimum latency and maximum efficiency. Figure 2 represents the workflow of the HMSA proposed.

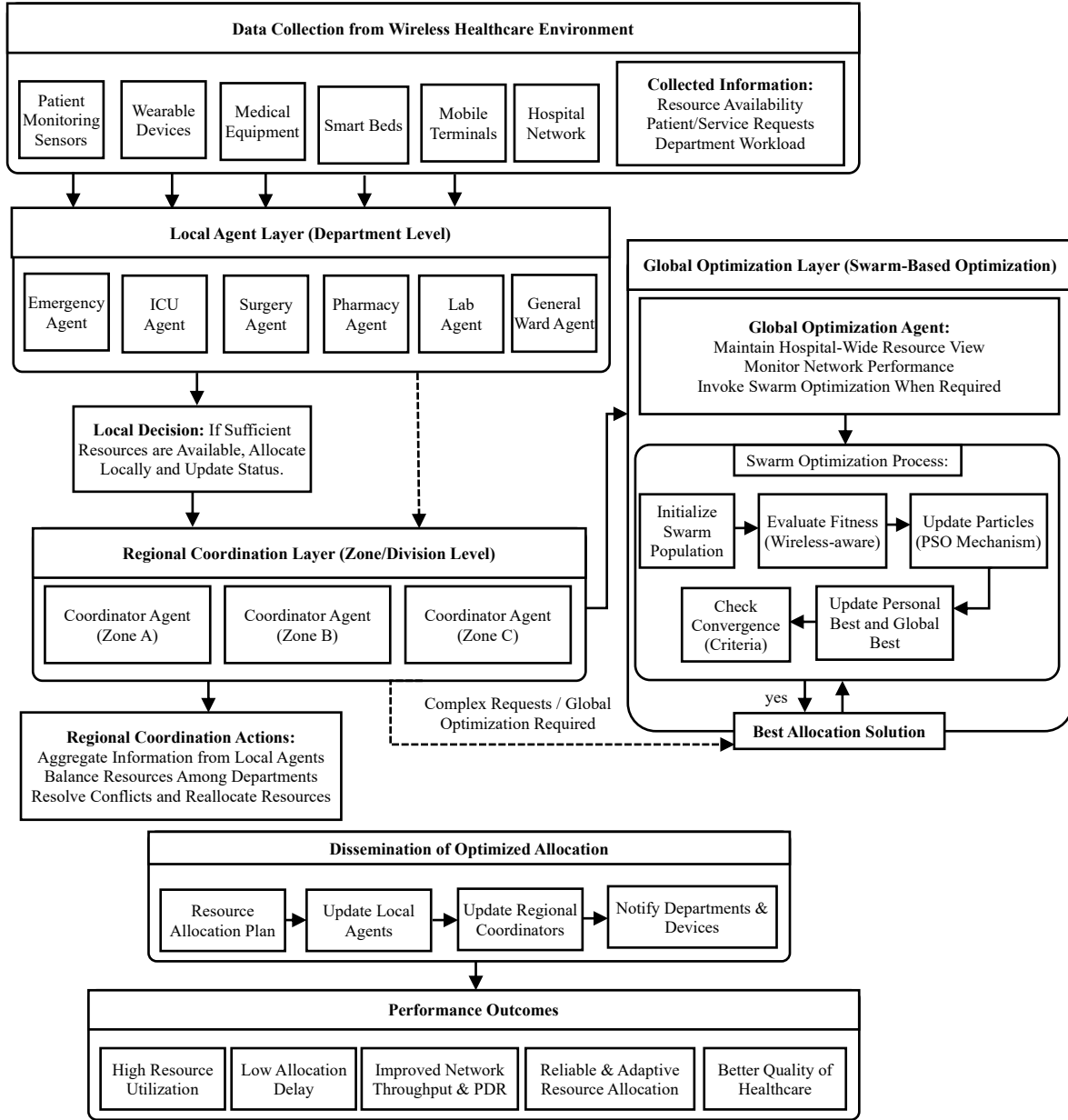


Figure 2: Workflow of the proposed hierarchical multi-agent swarm algorithm (HMSA)

4.1 Problem Formulation

Imagine a smart hospital context, which can be defined as a collection of healthcare resources. Let's represent it by equation 1

$$R = \{r_1, r_2, \dots, r_n\} \quad (1)$$

and a set of service requests represented as equation 2

$$T = \{t_1, t_2, \dots, t_m\} \quad (2)$$

The aim is to find an optimum strategy that will allocate the free resources available to the arriving demands and simultaneously provide high utilization and communication rates. The allocation decision is given by equations 3 and 4.

$$A = [a_{ij}] \quad (3)$$

Where

$$a_{ij} = \begin{cases} 1, & \text{if resource } r_i \text{ is assigned to task } t_j \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

The optimization objective seeks to maximize resource utilization while minimizing allocation delay and communication overhead.

4.2 Hierarchical Multi-Agent Coordination

The HMSA structure utilizes a three-level agent hierarchy: Local Agent, Regional Coordination Agent, and Global Optimization Agent. Local agents monitor departmental resources and make initial scheduling decisions. The Regional Coordination Agent is designed to manage resource sharing among multiple departments and allocate resources conflicting between different departments. The Global Optimization Agent is used to maintain a global overview of the hospital resources and conduct swarm optimization as soon as local and regional decision-making does not achieve a satisfactory outcome. This top-down coordination scheme reduces the computational complexity and enhances scalability of large hospital systems.

4.3 Swarm-Based Optimization Mechanism

The swarm optimization part views the candidate solution as a state of resource assignments. Each particle in a population searches the space for good solutions on resource assignments. The movement of a particle depends on personal and global best experiences, enabling the swarm to effectively search and exploit possible assignment decisions. The particle updates result in solutions for resource assignments that enhance global performance while adapting to varying hospital status.

The equations for the particle updates can be seen in the following equation 5.

$$\begin{aligned} V_i^{t+1} &= wV_i^t + c_1r_1(P_i - X_i^t) + c_2r_2(G - X_i^t) \\ X_i^{t+1} &= X_i^t + V_i^{t+1} \end{aligned} \quad (5)$$

Where V_i denotes velocity, X_i denotes particle position, P_i represents the personal best solution, and G denotes the global best solution.

4.4 Wireless-Aware Fitness Function

Communication restrictions existing in wireless healthcare networks have been addressed by HMSA through the inclusion of both operational and networking measurements in the optimisation problem. The fitness function checks resource utilization, allocation delay, network reliability, and communication efficiency.

The fitness function is given as equation 6.

$$F = \alpha U + \beta Q - \gamma D - \delta P \quad (6)$$

Where U denotes resource utilization, Q represents quality of service, D indicates allocation delay, and P corresponds to communication penalty factors associated with packet loss and network congestion. The weighting coefficients α , β , γ , and δ control the relative importance of each objective.

4.5 HMSA Algorithm

The design of HMSA is based on the initial gathering of available resources and services request data from healthcare devices and departmental agents. First, the local agents make initial allocation decisions based on departmental state. If the local resources are not enough, the requests will be sent to regional coordination agents to balance the resources among departments. In a complex allocation context, the Global Optimization Agent would activate the swarm optimization procedure to derive the hospital-wide optimal allocation strategy. The obtained allocation strategy is propagated among the agents, and it will be kept updated with the dynamic resource requirements and network state.

Algorithm 1: Hierarchical Multi-Agent Swarm Algorithm (HMSA)

Input: Resource set R , Task set T , Network status N

Output: Optimized resource allocation plan A

-
1. *Initialize Local Agents, Regional Agents, and Global Agent.*
 2. *Collect resource and task information from healthcare devices.*
 3. *Perform local resource allocation.*
 4. *Forward unresolved requests to Regional Agents.*
 5. *Execute inter-department coordination.*
 6. *Initialize swarm population.*
 7. *Evaluate fitness of candidate solutions.*
 8. *Update particle positions and velocities.*
 9. *Identify global best allocation strategy.*
 10. *Generate optimized allocation plan.*
 11. *Distribute allocation decisions throughout the hierarchy.*
 12. *Repeat optimization until convergence criteria are satisfied.*
-

HMSA leverages hierarchical agent coordination and swarm intelligence for dynamically adjusting its resource allocation decisions in response to changing operational needs and wireless network conditions. This enhances scalability, responsiveness, and overall healthcare service efficiency.

5 Experimental Design

A simulation was constructed to evaluate the efficiency of the suggested Hierarchical Multi-Agent Swarm Algorithm (HMSA), which simulates the operational activities of a large-scale smart hospital in a wireless health care network. IoMT devices, a wireless network system, several hospital departments, and real-time varied health care service requests were included in the simulation system. In this simulation, the proposed system's performance was designed to measure how efficiently HMSA can allocate resources with efficient communication and service quality under dynamic situations.

The simulated environment was built using the Python 3.10 programming language, where the development of the main algorithms was performed with the help of the NumPy and SciPy libraries for numerical computations and optimization. Simulations of the multi-agent interaction and event-based hospital process flow were made using the SimPy discrete event simulation library. Performance

characteristics of the network such as throughput, packet delivery ratio, and delay were simulated using the wireless communication simulation library. Data visualization, generation of graphs for performance comparisons, and one-way analysis of variance (ANOVA) tests were performed with the help of the Matplotlib and Seaborn libraries and SciPy.stats package, respectively. The simulation experiments were conducted on an Intel i7 computer with 16 GB RAM under the Ubuntu 22.04 LTS operating system.

5.1 Simulation Environment

A wireless hospital network comprising several departments such as the emergency department, intensive care unit, laboratory, pharmacy, and general wards; hospital beds and equipment; medical devices (monitoring devices); and human beings was set up in the simulation environment. Communication between wireless devices and agents occurred via an embedded IoMT network, and real-time data can be exchanged in that environment. The most important simulation parameters can be summarized in table 1.

Table 1: Simulation parameters

Parameter	Value
Number of Hospital Departments	15
Number of Hospital Beds	500
Medical Devices	300
IoMT Sensors	1,200
Local Agents	15
Regional Coordination Agents	3
Global Optimization Agent	1
Service Requests per Hour	500–3,000
Wireless Network Bandwidth	100 Mbps
Simulation Duration	24 Hours
Swarm Population Size	50
Maximum Iterations	100

5.2 Benchmark Algorithms

The efficiency of HMSA has been compared to that of some commonly adopted resource allocation methods. The benchmark methods considered are First-Come First-Served (FCFS), Round Robin (RR), Particle Swarm Optimization (PSO), and Ant Colony Optimization (ACO). These include simple scheduling techniques along with existing swarm intelligence-based optimization algorithms that have often been applied to distributed resource allocation problems.

5.3 Performance Metrics

The evaluation of the proposed system has been performed based on various operational and communication performance parameters: Resource Utilization (RU), defined as the percentage of usable resources employed on requests; Allocation Delay (AD), which shows the average time that it takes a request to receive a resource assigned; Network Throughput (NT), which reflects the volume of successful data delivery across the wireless network; Packet Delivery Ratio (PDR), defined as the communication performance measure, represents the ratio of packets delivered; Energy Consumption (EC) indicates the energy needed for communication and coordination tasks, whereas the QoS Satisfaction represents the percentage of requests fulfilled with the proposed requirements.

The metric for resource utilization is calculated using equation 7.

$$RU = \frac{R_{allocated}}{R_{available}} \times 100 \quad (7)$$

Where $R_{allocated}$ denotes the number of allocated resources and $R_{available}$ represents the total available resources.

The packet delivery ratio is calculated as equation 8

$$PDR = \frac{Packets_{received}}{Packets_{sent}} \times 100 \quad (8)$$

Where $Packets_{received}$ and $Packets_{sent}$ denote the number of received and transmitted packets, respectively.

5.4 Experimental Scenarios

Three simulation scenarios are presented to test the performance and robustness of HMSA in different operational contexts. The first scenario focuses on a standard hospital operation with limited-service requests and stable networks; the second one aims to study scalability by having an overwhelming number of patients admitted with enormous resource demand, and the third one attempts to investigate its performance in network congestions and volatile communication settings with its proposed wireless-aware optimization framework.

5.5 Statistical Validation

To confirm the stability and validity of the experimental results, all the simulations were conducted 30 times separately with the same parameters. The average and standard deviations for each of the performance measurements were calculated. A one-way ANOVA test at a significance level of 0.05 was employed to compare performance measurements of different algorithms. This procedure of verification assures that the performance gains obtained by HMSA are statistically significant rather than stochastic fluctuations in the environment.

6 Results and Discussion

6.1 Resource Utilization and Allocation Efficiency

In this study, HMSA was primarily assessed by its resource utilization and allocation efficiency. Table 2 gives a performance comparison of HMSA with respect to benchmark algorithms. As seen in table 2, HMSA has the maximum resource utilization (94.8%), which shows that it can discover and assign the most resources to the departments in a hospital. Moreover, it has the minimum allocation delay (96 ms), which benefits from hierarchical decision-making and distributed coordination. This result shows that local and regional agents could have minimized the resource conflicts; thus, the overall decision-making has been made at a fast pace after global optimization was called.

Table 2: Resource utilization and allocation delay comparison

Method	Resource Utilization (%)	Allocation Delay (ms)
FCFS	78.4	184
Round Robin	81.6	167
ACO	87.3	141
PSO	90.5	124
HMSA	94.8	96

Resource allocation reached by various resource allocation algorithms is shown in figure 3. It can be found that the proposed HMSA could achieve greater resource utilization over the existing scheduling and optimization algorithms by virtue of harmonizing resources at various organizational layers.

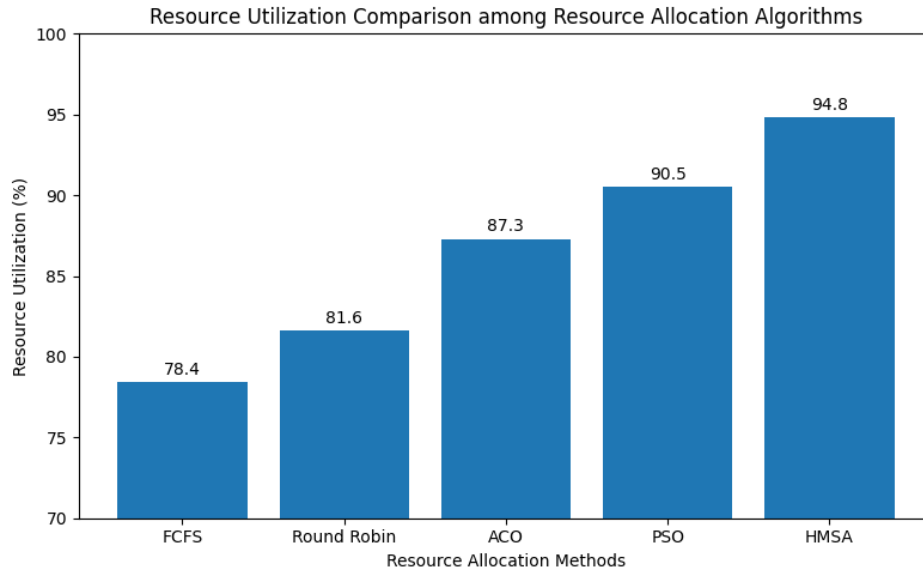


Figure 3: Resource utilization comparison among resource allocation algorithms

6.2 Wireless Network Performance Analysis

The performance of wireless communication is a key feature of a smart hospital environment, where allocation decisions rely on continuous information exchange between agents and health devices. Table 3 lists the throughput and packet delivery ratio for the allocation strategy under consideration.

Table 3: Wireless communication performance

Method	Throughput (Mbps)	Packet Delivery Ratio (%)
FCFS	76.5	90.2
Round Robin	79.8	91.5
ACO	85.6	94.1
PSO	89.7	96.2
HMSA	95.4	98.7

HMSA resulted in the largest throughput and packet delivery ratio. The communication-aware optimization enabled by the inclusion of the Network Monitoring Agent, which enables the system to adapt to network congestion, has improved transmission reliability. Higher throughput implies faster dissemination of resource allocation decisions and reliable transmission in healthcare communications. The throughput performance under various service request loads is shown in figure 4. Throughput is reduced when the load is high, but HMSA shows higher and reliable communications than the other schemes.

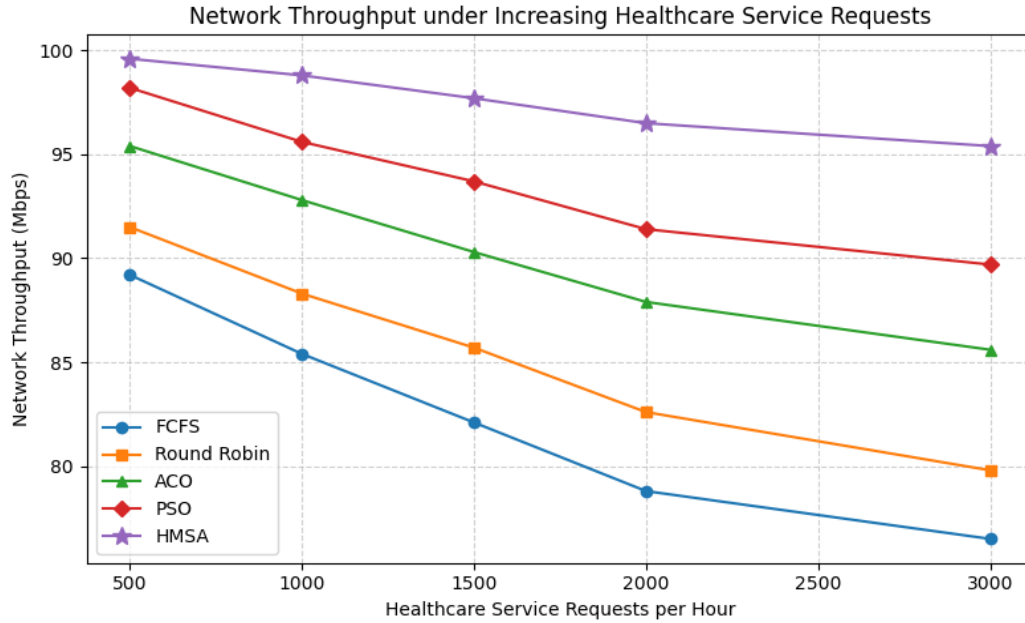


Figure 4: Network throughput under increasing healthcare service requests

6.3 Quality of Service and Scalability Evaluation

The effect of HMSA on the quality of service provided by healthcare services was also examined with QoS satisfaction and scalability analysis. Table 4 summarizes the achieved QoS satisfaction with a varied amount of service demand. For all methods, the performance degrades with an increasing number of service requests; however, HMSA still achieves much greater QoS satisfaction than rival methods.

Table 4: QoS satisfaction under different service request loads

Service Requests per Hour	FCFS (%)	Round Robin (%)	ACO (%)	PSO (%)	HMSA (%)
500	90.8	92.4	95.6	97.2	99.1
1000	88.1	90.5	93.8	96.4	98.3
1500	85.9	88.7	92.1	95.2	97.6
2000	82.7	85.9	89.4	93.1	96.8
3000	79.5	82.4	86.7	90.6	95.1

The better QoS can be attributed to the dynamic nature of the hierarchical agent structure in resource redistribution on request. HMSA proves to perform better when under a huge traffic load, as shown by the results, which is applicable for large-scale healthcare settings.

6.4 Statistical Validation

Table 5: One-Way ANOVA results for resource utilization

Source of Variation	Sum of Squares (SS)	Degrees of Freedom (df)	Mean Square (MS)	F-value	p-value
Between Groups	924.36	4	231.09	32.84	< 0.001
Within Groups	1020.15	145	7.04	—	—
Total	1944.51	149	—	—	—

The results of the one-way ANOVA are displayed in table 5 and demonstrate that the considered resource allocation algorithms have significantly different results. The value of F 32.84 is greater than the critical value for a 95% confidence level, and the value of p ($32.84 < 0.001$) indicates that the differences observed in resource usage are extremely significant. The values demonstrate that the increase in resource efficiency by HMSA is not as a result of coincidence but is indeed due to the efficacy of the suggested hierarchical multi-agent swarm optimization method. The statistical analysis has therefore validated the fact that HMSA is a more efficient approach than FCFS, Round Robin, ACO, and original PSO when allocating hospital resources.

7 Discussion

The experimental results demonstrate that the proposed HMSA framework significantly improves hospital resource allocation performance compared with conventional scheduling and optimization methods. The superior resource utilization and reduced allocation delay achieved by HMSA can be attributed to its hierarchical coordination mechanism, which distributes decision-making responsibilities across local, regional, and global agents. This structure enables faster responses to resource requests while reducing computational bottlenecks associated with centralized allocation approaches. Furthermore, the swarm optimization component effectively explores multiple allocation alternatives and identifies near-optimal resource assignment strategies, resulting in improved operational efficiency and better utilization of available healthcare resources.

In addition to operational improvements, HMSA exhibits strong communication performance within wireless healthcare environments. The integration of network-aware parameters into the optimization process enables the framework to maintain high throughput, packet delivery ratio, and QoS satisfaction even under increasing service demand. The results indicate that HMSA can adapt to dynamic workload variations while preserving communication reliability, which is essential for real-time healthcare applications. The statistical validation further confirms that the observed performance gains are significant and consistent across repeated simulation runs. Therefore, the proposed HMSA framework provides an effective and scalable solution for intelligent hospital resource allocation in next-generation wireless healthcare networks.

Limitations and Future Work

Although the proposed HMSA framework demonstrated significant improvements in resource utilization, allocation delay, throughput, and QoS performance, several limitations should be acknowledged. First, the evaluation was conducted in a simulation-based environment, which may not fully capture the complexity and unpredictability of real-world hospital operations. Second, the study assumes reliable agent communication and does not explicitly consider severe network failures, cyberattacks, or hardware malfunctions that may occur in practical healthcare deployments. Third, the optimization model focuses primarily on resource allocation and wireless communication performance, while factors such as patient prioritization policies, clinical workflows, and human decision-making behaviors were not incorporated. Additionally, the computational overhead associated with large-scale agent coordination may increase as the size of the healthcare network grows substantially.

Future research will focus on validating the proposed HMSA framework in real hospital environments and integrating advanced intelligent technologies to enhance decision-making capabilities. The incorporation of federated learning can enable privacy-preserving collaborative optimization across multiple healthcare facilities, while digital twin technologies can provide real-time

virtual representations of hospital operations for predictive resource management. Furthermore, blockchain-based security mechanisms may be integrated to strengthen data integrity, trust management, and secure agent communication. The adoption of emerging 6G-enabled healthcare networks and edge intelligence frameworks can further improve scalability, ultra-low-latency communication, and autonomous resource orchestration in next-generation smart healthcare systems.

8 Conclusion

This study presented a Hierarchical Multi-Agent Swarm Algorithm (HMSA) for optimized hospital resource allocation over wireless healthcare networks. The proposed framework combines hierarchical multi-agent coordination and swarm intelligence optimization to address the challenges associated with dynamic healthcare demands, heterogeneous medical resources, and communication constraints in smart hospital environments. By integrating Healthcare Device, Local Agent, Regional Coordination, and Global Optimization layers, the framework enables distributed decision-making while maintaining efficient resource utilization and communication performance. Furthermore, the incorporation of wireless network metrics into the optimization process allows the system to adapt allocation decisions according to prevailing network conditions, thereby improving overall dependability and service quality. Comprehensive simulation experiments demonstrated the effectiveness of HMSA when compared with FCFS, Round Robin, ACO, and conventional PSO approaches. The proposed method achieved the highest resource utilization (94.8%), throughput (95.4 Mbps), packet delivery ratio (98.7%), and QoS satisfaction (97.1%), while producing the lowest allocation delay (96 ms). These improvements highlight the capability of HMSA to efficiently coordinate healthcare resources and maintain reliable wireless communication under varying operational conditions. The hierarchical coordination mechanism significantly reduced resource contention and scheduling overhead, whereas the swarm optimization component enabled adaptive and near-optimal allocation decisions. The statistical analysis further confirmed the robustness of the proposed framework. One-way ANOVA results yielded an F-value of 32.84 and $p < 0.001$, indicating statistically significant differences among the evaluated algorithms at the 95% confidence level. Therefore, HMSA represents a promising solution for next-generation smart hospitals and wireless healthcare systems. Future research will focus on integrating federated learning, digital twins, blockchain-enabled security mechanisms, and 6G-assisted healthcare communication infrastructures to further enhance resource management efficiency and system intelligence.

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